

POINTE DE CHEMOULIN, France

WMO: **072160**Lat: **47.2340N**Lon: **2.2989W**Elev: **20**StdP: **101.08**Time Zone: **1.00 (EUC)**Period: **01-14**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.5	-0.1	-4.9	2.5	0.7	-3.2	2.9	1.8	25.3	9.6	22.5	9.8	4.8	40	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	28.2	20.2	26.0	19.4	24.2	18.3	21.0	26.0	20.1	24.3	19.3	22.8	4.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.2	22.8	18.6	13.5	21.9	17.9	12.9	21.1	60.7	25.7	57.4	24.6	55.0	23.1	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	16.4	13.9	-3.5	32.2	1.7	2.2	-4.7	33.8	-5.7	35.1	-6.6	36.3	-7.8	37.9		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	7.2	7.3	9.5	12.1	14.6	17.8	19.3	18.9	17.8	14.9	10.7	7.6
	DBStd	5.23	3.06	2.90	2.82	2.70	2.52	2.65	2.45	2.08	2.29	2.58	2.88	3.24
	HDD10.0	334	91	79	42	8	0	0	0	0	0	3	26	86
	HDD18.3	2014	344	309	274	188	119	41	15	15	37	108	231	333
	CDD10.0	1491	4	3	26	71	144	234	288	277	233	154	46	11
	CDD18.3	131	0	0	0	1	4	26	45	34	19	2	0	0
	CDH23.3	577	0	0	0	8	30	122	226	122	66	3	0	0
	CDH26.7	135	0	0	0	3	3	26	64	30	8	1	0	0
Wind	WSAvg	6.0	6.4	6.3	5.8	5.8	6.3	5.7	5.9	5.6	5.1	6.3	6.1	7.1
Precipitation	PrecAvg	682	75	60	47	50	48	33	37	36	53	80	84	78
	PrecMax	906	156	129	128	130	97	66	79	96	138	170	174	150
	PrecMin	460	9	8	4	12	7	4	3	1	2	20	21	3
	PrecStd	107	39	34	31	29	22	18	18	25	35	37	43	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	14.0	21.3	23.0	26.7	29.7	31.2	31.1	27.8	22.3	17.3	14.3
		MCWB	11.7	10.3	15.6	16.1	18.8	21.0	19.8	22.4	20.2	17.9	15.0	12.9
	2%	DB	12.0	12.1	17.2	20.1	23.2	26.5	28.5	27.0	25.1	20.1	16.2	13.2
		MCWB	11.2	10.1	12.4	14.4	17.6	19.7	19.6	20.2	17.9	16.6	13.9	11.9
	5%	DB	11.4	11.3	14.7	18.1	20.8	24.3	25.8	24.3	23.4	18.9	15.3	12.4
		MCWB	10.7	9.9	10.7	13.7	16.2	18.5	18.9	18.8	17.1	16.3	13.3	11.3
	10%	DB	10.9	10.7	13.0	16.4	18.7	22.4	23.4	22.5	21.7	18.1	14.6	11.8
		MCWB	10.2	9.6	10.4	12.9	15.0	18.0	18.5	18.3	16.7	16.1	12.7	10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	11.6	14.3	16.7	19.5	21.7	21.2	23.3	20.7	18.2	15.7	13.3
		MCDB	12.6	12.5	17.8	20.0	25.3	27.5	27.5	29.9	25.5	20.1	16.8	13.8
	2%	WB	11.7	10.9	12.4	15.4	18.0	20.7	20.2	21.4	19.3	17.4	14.5	12.2
		MCDB	12.1	11.4	14.6	18.1	22.6	25.3	25.5	25.9	22.5	18.9	15.6	12.7
	5%	WB	11.0	10.5	11.6	14.1	16.9	19.5	19.6	20.1	18.6	16.9	13.6	11.5
		MCDB	11.2	11.1	12.9	16.7	20.3	23.4	23.9	23.2	21.1	18.0	14.7	12.1
	10%	WB	10.4	10.0	11.0	13.1	15.7	18.5	19.0	19.0	17.9	16.4	12.7	10.8
		MCDB	10.8	10.6	12.0	15.7	18.2	21.8	22.4	22.0	20.3	17.5	13.8	11.6
Mean Daily Temperature Range	5% DB	MDBR	3.8	4.8	6.0	6.8	6.6	7.2	6.3	6.4	7.7	5.0	5.0	4.4
		MCDBR	3.3	5.1	9.5	10.5	10.8	11.5	11.9	10.3	10.7	6.2	5.2	3.6
		MCWBR	2.8	3.9	5.9	5.9	6.1	5.8	5.6	5.5	5.5	4.3	4.2	3.3
	5% WB	MCDBR	2.5	3.5	6.0	8.3	9.2	9.8	8.6	9.0	7.4	4.9	3.8	3.0
		MCWBR	2.6	3.1	4.2	5.3	5.5	5.3	4.4	4.9	4.7	3.6	3.6	3.3
Clear-Sky Solar Irradiance	taub	0.334	0.348	0.376	0.391	0.396	0.399	0.394	0.383	0.367	0.365	0.341	0.328	
	taud	2.430	2.405	2.334	2.300	2.334	2.358	2.374	2.422	2.446	2.444	2.442	2.410	
	Ebn at Noon	743	813	841	862	869	866	865	862	844	782	727	700	
	Edh at Noon	69	86	107	121	122	120	116	106	95	82	68	63	
All-Sky Solar Radiation	RadAvg	1.09	2.03	3.26	4.84	5.72	6.28	6.01	5.29	4.20	2.41	1.38	0.96	
	RadStd	0.10	0.20	0.34	0.47	0.39	0.54	0.27	0.37	0.28	0.19	0.12	0.11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page